

Bayu Nismara Nagatama

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ABOUT

Motivated and detail-oriented Informatics Engineering student at Institut Teknologi Sepuluh Nopember (ITS) Surabaya, currently in my 6th semester, with a strong focus on front-end web development. I have hands-on professional experience through a full-time internship at Roundbytes and an ongoing contract role as a Front-End Developer at PLN Group, where I have built and maintained production-level web applications using modern frameworks including Vue.js 3, TypeScript, and Tailwind CSS. Beyond professional experience, I actively apply deep learning and AI concepts in academic research, including semantic segmentation on medical imaging data for my undergraduate thesis. I also have a track record in competitions, earning regional and international recognition in technology innovation and the arts. I thrive in collaborative, fast-paced environments and am eager to contribute meaningful, clean, and scalable front-end solutions while continuing to grow as a well-rounded software developer.

EDUCATION

Institut Teknologi Sepuluh Nopember (ITS)

Aug 2023 — Present

Bachelor's Student of Informatics Engineering (Teknik Informatika) — GPA: 3.64 / 4.00

- Currently in 6th semester; active in academic projects spanning front-end web development, deep learning, and AI-assisted application design.
- Relevant coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Systems, Human-Computer Interaction, Machine Learning, Web Programming.
- Undergraduate thesis: Semantic segmentation of dental CBCT images using YOLO-based architecture to classify alveolar bone quality (Lekholm & Zarb D1–D4), with evaluation metrics DSC, IoU, HD95, Accuracy, Precision, Recall, and F1.

WORK EXPERIENCE

PLN Group — Front-End Web Developer (Contract)

Oct 2025 — Present

PT PLN (Persero) — Surabaya, Indonesia

- Developed the web-based user interface for an internal Legal AI Assistant, enabling staff to query and navigate legal documents through a clean, accessible UI.
- Designed and built a premium, responsive landing page and dashboard using Vue.js 3 and Tailwind CSS, ensuring cross-browser compatibility and mobile responsiveness.
- Collaborated closely with AI/ML and backend teams in an enterprise environment, translating system requirements and API contracts into polished front-end experiences.
- Maintained code quality through component-based architecture and consistent design system adherence across multiple internal tools.

Roundbytes — Full-Time Web Developer Intern

Apr 2024 — Aug 2024

Roundbytes — Surabaya, Indonesia

- Designed, developed, and maintained client websites using WordPress, delivering responsive and visually consistent digital products.
- Coded responsive web pages using TypeScript and Tailwind CSS, improving page performance and maintainability for multiple client projects.
- Collaborated with the development team to deliver on client requirements, participating in sprint planning and code reviews.
- Gained practical experience in modern front-end web technologies and deployment practices in a professional agency setting.

RESEARCH & PROJECTS

Undergraduate Thesis — Dental Alveolar Bone Quality Classification

2025 — Present

ITS — Supervised Tugas Akhir Research

- Researching YOLO-based semantic segmentation on CBCT dental images to classify alveolar bone quality under the Lekholm & Zarb D1–D4 grading system.
- Defined cortical-to-trabecular ratio metric and a full evaluation framework including DSC, IoU, HD95, Accuracy, Precision, Recall, and F1 Score.
- Authored a full bilingual proposal (Indonesian & English abstract) covering literature review, pseudocode-level methodology, comparative literature gap analysis, and Gantt chart.

PharmaCare+ — AI-Powered Medication Adherence Platform

Oct — Dec 2025

Competition Project — Inovasi Surabaya (INOVBOYO) 2025

- Developed a digital platform helping elderly users and caregivers improve medication adherence through automated reminders, dangerous drug-interaction detection, and online pharmacy consultation services.

- Built using AI-driven features for real-time drug interaction screening and notification workflows; deployed as a working web application (pharmacare-gericommunity.lovable.app).
- Competed in the INOVBOYO 2025 regional innovation competition organized by Badan Riset dan Inovasi Daerah Provinsi Jawa Timur, achieving 2nd Runner-Up (Juara Harapan 2).

VISULEARN AI — AI-Powered Visual Learning Card System

2025

Competition Project — Samsung Solve for Tomorrow 2026

- Designed an AI pipeline converting teacher speech into five structured visual card templates (Process, Comparison, Hierarchy, Timeline, Concept) tailored for Deaf students in Indonesian SMP/SMA science (IPA) classes.
- Integrated Whisper STT with a curated academic-to-plain-language dictionary and teacher validation gate to maintain pedagogical control over AI-generated outputs.
- Produced full system architecture document and competition proposal paper.

SkillBridge — Career Matching & Competency Gap Analyzer (Capstone Project)

Feb — Jun 2026

Industry Capstone Project with PT Sentra Vidya Utama (SEVIMA) — Front-End Developer, 7-member team

- Built the Laravel 12 + Tailwind CSS v4 front-end for an AI-powered platform that extracts skills from CVs via Claude Sonnet 4.6, analyzes skill gaps against 5 IT role standards, and delivers MCQ-based skill verification and learning path recommendations.
- Implemented the authentication proxy layer (login/register/logout) bridging Laravel sessions with the FastAPI backend's JWT tokens, including custom ApiAuth and ApiGuest middleware for route protection.
- Developed core user-facing Blade views covering the CV upload, skill management, target role selection, skill gap visualization, and learning path pages, consuming a centralized browser-side API client.
- Collaborated with backend, ML, and QA sub-teams in an agile workflow with sprint reviews, contributing to a system that passed 55 backend unit tests, 8 frontend unit tests, and 23 integration tests.

TENUNA: The Living Threads — Interactive Editorial Documentary Website

2026

Cultural Digital Initiative — Competition Project (BytesFest 2026)

- Built an interactive editorial documentary website preserving and celebrating Sumba Ikat weaving heritage, aligned with SDG 11 (Sustainable Cities and Communities), using React 19, TypeScript, and Vite 8.
- Implemented scroll-driven storytelling with GSAP and ScrollTrigger, synchronizing text transitions, parallax effects, and a custom animated SVG 'Global Thread' element that visually evolves in thickness, color, and pattern as the user scrolls.
- Developed a centralized bilingual (Indonesian/English) system using React Context API, paired with an interactive drag-and-flip language switcher built with Framer Motion for physics-based micro-interactions.
- Built a 3D WebGL circular gallery (OGL) for interactive motif exploration and a Bento Grid layout for philosophical motif storytelling, alongside performance optimizations (WebP compression, layout-shift elimination) for instant First Contentful Paint.
- Diagnosed and resolved layout stability issues, including footer hover jitter (fixed-height containers) and z-index layering conflicts between background visuals, animated SVG elements, and foreground text.

ORGANIZATIONAL EXPERIENCE

Staff — Departemen Rumah Tangga, UKM PSM ITS

Mar 2025 — Feb 2026

Unit Kegiatan Mahasiswa Paduan Suara Mahasiswa — ITS, Surabaya

- Served in the internal affairs and family department of ITS's student choir organization, handling organizational welfare and internal communications.
- Acted as PIC (Person in Charge) for the procurement and ordering process of the official organization jacket, coordinating between members and vendors end-to-end.

Head of Operational Division — Organizing Committee, UKM PSM ITS

Jul 2024 — Jul 2025

Unit Kegiatan Mahasiswa Paduan Suara Mahasiswa — ITS, Surabaya

- Led the Operational Division for a year-long new member cadre program (kaderisasi), coordinating closely with other divisions to ensure the event ran smoothly from planning through execution.
- Managed all operational aspects including vendor coordination, venue arrangements, and logistics planning.
- Assigned and delegated tasks to division staff, ensuring accountability and timely completion of responsibilities.
- Ensured the needs of both partner divisions and program participants were consistently met throughout the year-long program.

Head of Regular Group — Dewan Ambalan (Scout Leadership)

Aug 2020 — Jun 2023

Frateran High School — Surabaya, Indonesia

- Led a scout group across three years, coordinating regular programs, events, and leadership development activities for members.

Staff — Event Planner & Commentator, Frateran Cup: Ace of Sports

Oct 2021

Frateran High School — Surabaya, Indonesia

- Organized and hosted a school-wide sports event; managed logistics, scheduling, and live commentary for multiple concurrent sport competitions.

ACHIEVEMENTS

Gold Medal (x3) & 1st Place (Folklore) — 3rd Soegijapranata International Choir Competition
Soegijapranata Catholic University — International Competition

Nov 2024

- Competed as part of ITS's choir ensemble across three categories: Folklore, Sacred, and Mixed Voice.
- Won a Gold Medal in all three categories; achieved 1st Place overall in the Folklore category.

2nd Runner-Up (Juara Harapan 2) — Inovasi Surabaya (INOVOYO) 2025
Badan Riset dan Inovasi Daerah Provinsi Jawa Timur — Regional Competition

Oct — Dec 2025

- Awarded 2nd Runner-Up in the INOVOYO 2025 regional innovation competition for PharmaCare+, an AI-powered medication adherence platform for the elderly.

Semifinalist — Ubaya ILPC 2023 (Programming & Logic Competition)
Universitas Surabaya — Surabaya, Indonesia

Feb 2023

- Reached the semifinal stage in a regional competitive programming and logic reasoning competition.

2nd Place (Regional) — Bebras Ubaya Region (Logic Competition)
Ubaya Region — Surabaya, Indonesia

Nov 2021

- Achieved 2nd place in the Ubaya regional round of the international Bebras computational thinking competition.

CERTIFICATIONS

- Build an AI Agent — IBM SkillsBuild (AI Agents, IBM watsonx.ai, Agentic AI, Generative AI, Evaluation Metrics, AI Ethics)
- IT — AI Agent for Programming — IBM SkillsBuild University Education (hosted by Hacktiv8 Indonesia)
- AWS Academy Cloud Security Foundations — Amazon Web Services (AWS, AWS Cloud, AWS Cloud Security, Cloud Security)
- Palo Alto Cybersecurity Foundation — Palo Alto Networks

SKILLS

Front-End Development: Vue.js 3, React, TypeScript, JavaScript, HTML/CSS, Tailwind CSS, GSAP, Framer Motion, WordPress

Programming Languages: Python, JavaScript, TypeScript, SQL, C++

AI / Deep Learning: PyTorch, TensorFlow, Scikit-learn, OpenCV, YOLO, Whisper STT, IBM watsonx.ai

Tools & Platforms: Git/GitHub, VS Code, Google Colab, Jupyter Notebook, Microsoft Office

Operating Systems: Windows, Linux (Ubuntu)

Soft Skills: Critical Thinking, Team Collaboration, Time Management, Technical Writing, Problem Solving

Languages: Bahasa Indonesia (Native), English — Upper-Intermediate B2